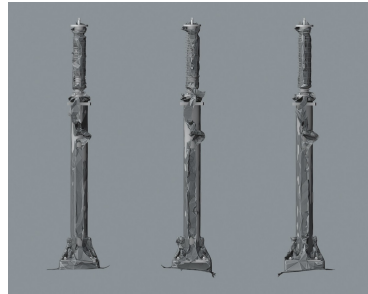


Why the Reconstruction Loop Broadens: Three Structural Transitions

Each row shows how a new dependency changes the engineering decision and therefore the operator choice.

A · Fit →
reusable
scope

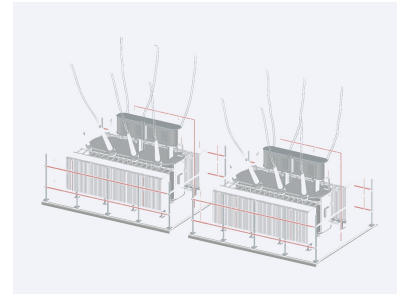
B01: local fit



define a comparison domain;
fit pose and size

repetition
→
invariance

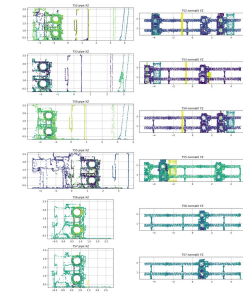
B08: shared equipment



two installations reuse one
shared MASTER

reuse
boundary
testable

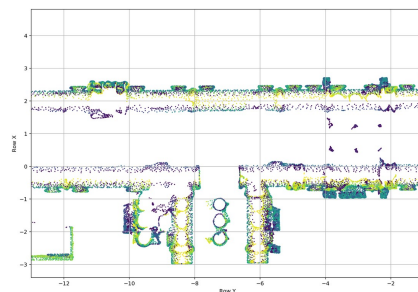
B15: revise reuse boundary



finite site geometry moves
outside the shared asset

B · Object →
connected
representation

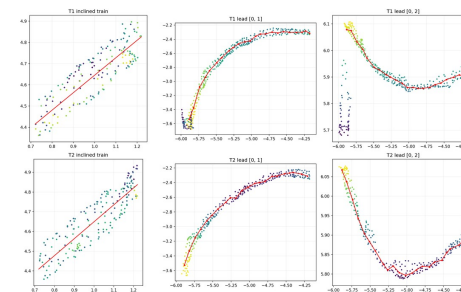
B20: ports and routes



existing equipment ends
become explicit ports

relationships
appear

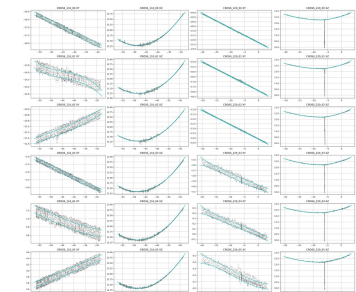
B23: change representation



connection reformulated:
straight → curved

endpoint
constraints
scale

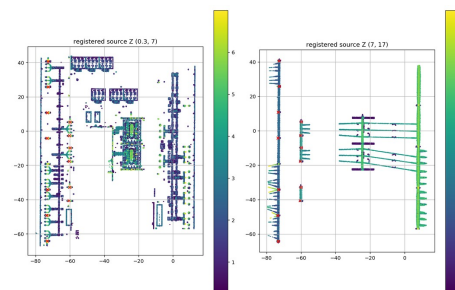
B29: relational constraints



continuity constraints scale
across subsystems

C · Addition
→ state-
driven
closure

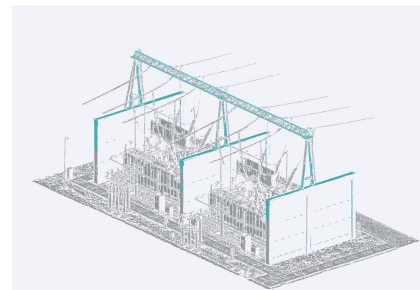
B25: unexplained geometry



large unexplained structures
remain in the scene

coverage →
task
selection

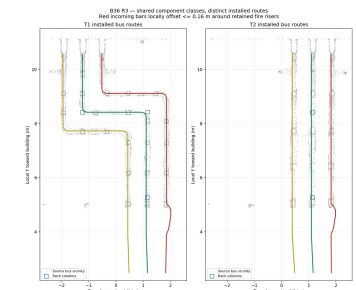
B26: reconstruct selected gaps



coverage selects what to
reconstruct next

fit the
inherited
world

B36: state-constrained insertion



new geometry must fit
inherited endpoints

Descriptive evidence from the B01–B36 record. The figure explains how the reconstruction problem changes; it does not propose a universal complexity taxonomy.